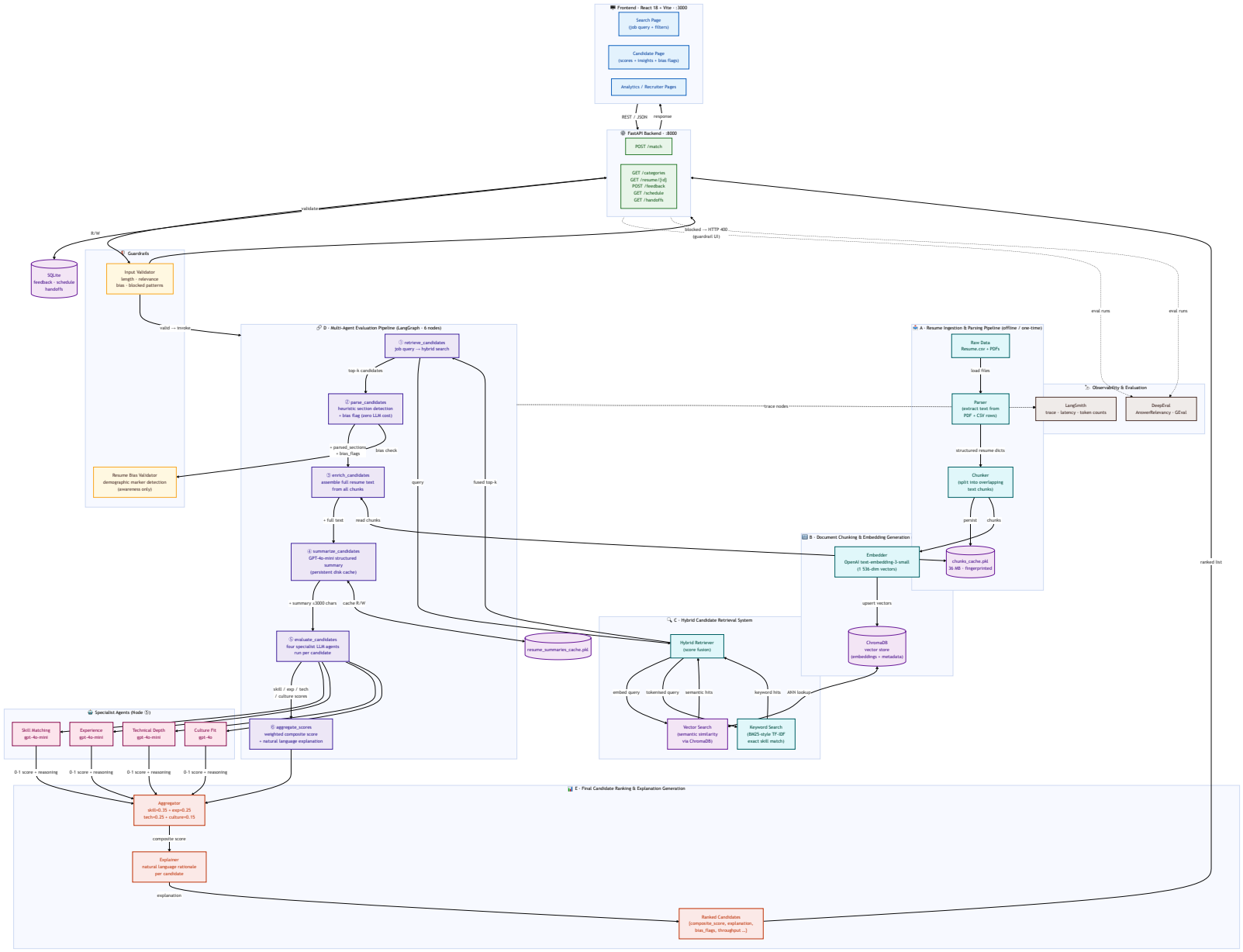
Frontend · React 18 + Vite · :3000
Search Page
(job query + filters)
Candidate Page
(scores + insights + bias flags)
Analytics / Recruiter Pages
REST / JSON
response
FastAPI Backend · :8000
POST /match
GET /categories
GET /resume/{id}
POST /feedback
GET /schedule
GET /handoffs
validate
R/W
blocked → HTTP 400
(guardrail UI)
SQLite
feedback · schedule
handoffs
Guardrails
Input Validator
length · relevance
bias · blocked patterns
valid → invoke
Resume Bias Validator
demographic marker detection
(awareness only)
D · Multi-Agent Evaluation Pipeline (LangGraph · 6 nodes)
① retrieve_candidates
job query → hybrid search
top-k candidates
② parse_candidates
heuristic section detection
+ bias flag (zero LLM cost)
+ parsed_sections
+ bias_flags
bias check
③ enrich_candidates
assemble full resume text
from all chunks
+ full text
read chunks
④ summarize_candidates
GPT-4o-mini structured
summary
(persistent disk cache)
+ summary x3000 chars
cache R/W
⑤ evaluate_candidates
four specialist LLM agents
run per candidate
skill / exp / tech
/ culture scores
⑥ aggregate_scores
weighted composite score
+ natural language explanation
query
fused top-k
resume_summaries_cache.pkl
trace nodes
A · Resume Ingestion & Parsing Pipeline (offline / one-time)
Raw Data
Resume.csv + PDFs
load files
Parser
(extract text from
PDF + CSV rows)
structured resume dicts
Chunker
(split into overlapping
text chunks)
persist
chunks
chunks_cache.pkl
36 MB · fingerprinted
Observability & Evaluation
LangSmith
trace · latency · token counts
DeepEval
AnswerRelevancy · GEval
eval runs
eval runs
B · Document Chunking & Embedding Generation
Embedder
OpenAI text-embedding-3-small
(1 536-dim vectors)
upsert vectors
ChromaDB
vector store
(embeddings + metadata)
C · Hybrid Candidate Retrieval System
Hybrid Retriever
(score fusion)
embed query
tokenised query
semantic hits
keyword hits
ANN lookup
Vector Search
(semantic similarity
via ChromaDB)
Keyword Search
(BM25-style TF-IDF
exact skill match)
Specialist Agents (Node ⑤)
Skill Matching
gpt-4o-mini
Experience
gpt-4o-mini
Technical Depth
gpt-4o-mini
Culture Fit
gpt-4o
0-1 score + reasoning
0-1 score + reasoning
0-1 score + reasoning
0-1 score + reasoning
E · Final Candidate Ranking & Explanation Generation
Aggregator
skill=0.35 + exp=0.25
tech=0.25 + culture=0.15
composite score
Explainer
natural language rationale
per candidate
explanation
Ranked Candidates
[composite_score, explanation,
bias_flags, throughput ...]
ranked list